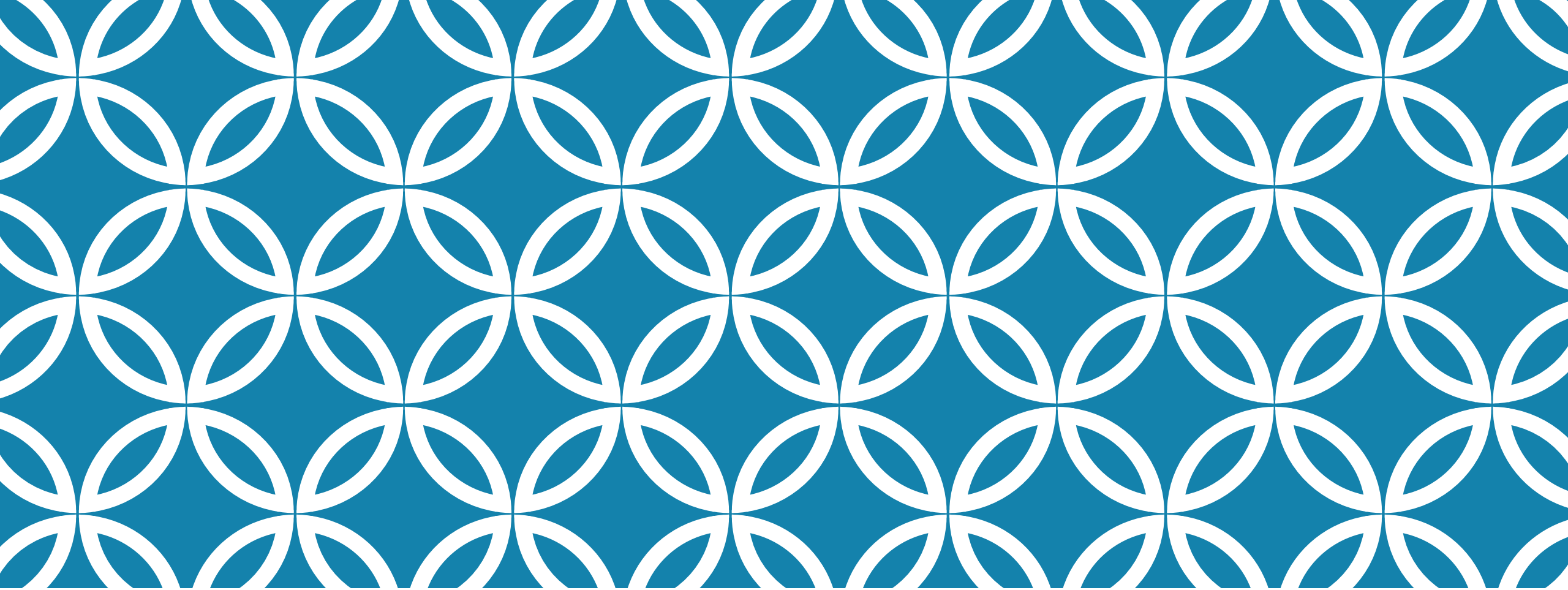


DATA GOVERNANCE, COMPLIANCE AND ETHICS

WEEK 02: DATA GOVERNANCE AND MANAGEMENT (PART 2)

jaswinder.singh@ncirl.ie

OVERVIEW OF COURSE CONTENT

1. Data Governance and Management (Weeks 1-2)
2. Privacy and Data Protection (Weeks 3-5)
3. Ethical Issues Pertaining to Data (Weeks 6, 8)
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 9-12)

* Week 7 = Reading Week (No classes)

AGENDA

Week 1: Introduction to Data Governance (DG)

- Definitions of DG
- Data Governance & Data Management
- Good and Poor DG
- Introduction to DMBOK wheel

Week 2 : Application of concept

- Data Management principles
- Management of data strategies, policies, processes and procedures
- Data Quality, Security and Integrity
- Roles and responsibilities

Supplementary Material

- Data risk identification and management
- Models and tools for data governance

OVERVIEW OFCOURSE CONTENT

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-5)
3. Ethical Issues Pertaining to Data (Weeks 6-8)*
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 9-12)

IMPORTANCE OF ASSIGNING ROLES AND RESPONSIBILITIES TO DG

- "Like any other activity within a company or government entity, there needs to be a **formal statement of roles**.
- The **official designation of accountability and responsibility** are key factors to the survival of DG.
- In many organizations, the responsible parties have a formal role as designated **"stewards" or "custodians."**

- The organization around DG also requires a **hierarchy** of some sort to **enable issue resolution, monitoring, and direction setting**.
- Rarely does this hierarchy of DG become a stand-alone area (i.e., **there is rarely a data governance "department"**).
- Most of the time, the DG organization is a **virtual organization** made up of business and IT personnel.

Chapter 3. Overview of a data governance program | Data Governance (oreilly.com)

DG ROLES AND RESPONSIBILITIES (1/2)

- Roles and Responsibilities (Rs & Rs) are the backbone of a successful DG program
- Rs & Rs formalize accountability of how people define, produce, and use data
- Roles should be defined and deployed according to the culture and structure (size, type) of the organization (DG operating model)

- In a larger organization, you might find a range of roles that are business, IT, or data management specific:
  - Chief Data Officer
  - Data Protection/Privacy Officer (DPO)
  - Data Steward
  - Data Architect
  - Chief Risk Officer
  - Chief Analytics Officer

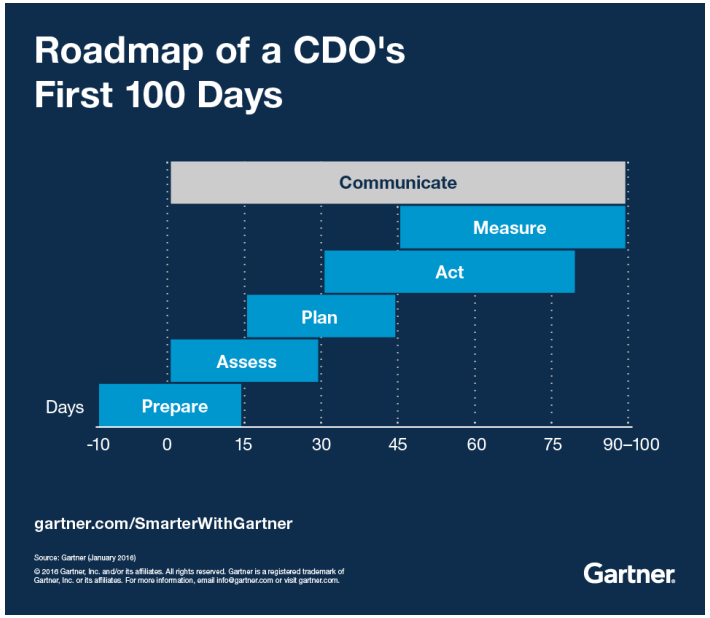


DG ROLES AND RESPONSIBILITIES (2/2)

Data Steward Responsible	A Data Steward is a person that defines, produces or uses data as part of their job and has a defined level of responsibility for assuring quality in the definition, production or usage of that data. Data Stewards responsibilities include: • Developing and maintaining data classification policies. • Developing, implementing, and managing data access policies. • Ensuring that data quality and data definition standards are developed and implemented. • Resolving stewardship issues and data definitions of data elements that cross multiple functional units.
Data Trustee Accountable	Data Trustees are defined as institutional officers, (i.e. Vice Presidents, Vice Provosts, Deans, Chancellors, etc.) who have authority over policies and procedures regarding business definitions of data, and the access and usage of that data, within their delegations of authority. Each Data Trustee appoints Data Stewards for their specific Subject Area Domains.
Data Custodian Supportive	Data Custodians are system administrators responsible for the operation and management of systems and servers which collect, manage, and provide access to institutional data. Data Custodian responsibilities include: • Maintaining physical and system security and safeguards appropriate to the classification level of the data in their custody. • Maintaining Disaster Recovery plans and facilities appropriate to business needs and adequate to maintain or restart operations in the event systems or facilities are impaired, inaccessible, or destroyed. • Managing Data User access as prescribed and authorized by appropriate Data Stewards. • Following data handling and protection policies and procedures established by appropriate Data Stewards.
Subject Matter Expert Consulted	A subject-matter expert (SME) are those individuals that support and consult the business and the technical professionals with their knowledge of business operations and the data that is necessary to operate and perform analysis. These people can be Business Analysts, Reporting Analysts, Data Architects, Data Modelers, and Project Management.
Data Users Informed	Data users are university units or individual university community members who have been granted access to institutional data in order to perform assigned duties or in fulfillment of assigned roles or functions within the university; this access is granted solely for the conduct of university business

CHIEF DATA OFFICER

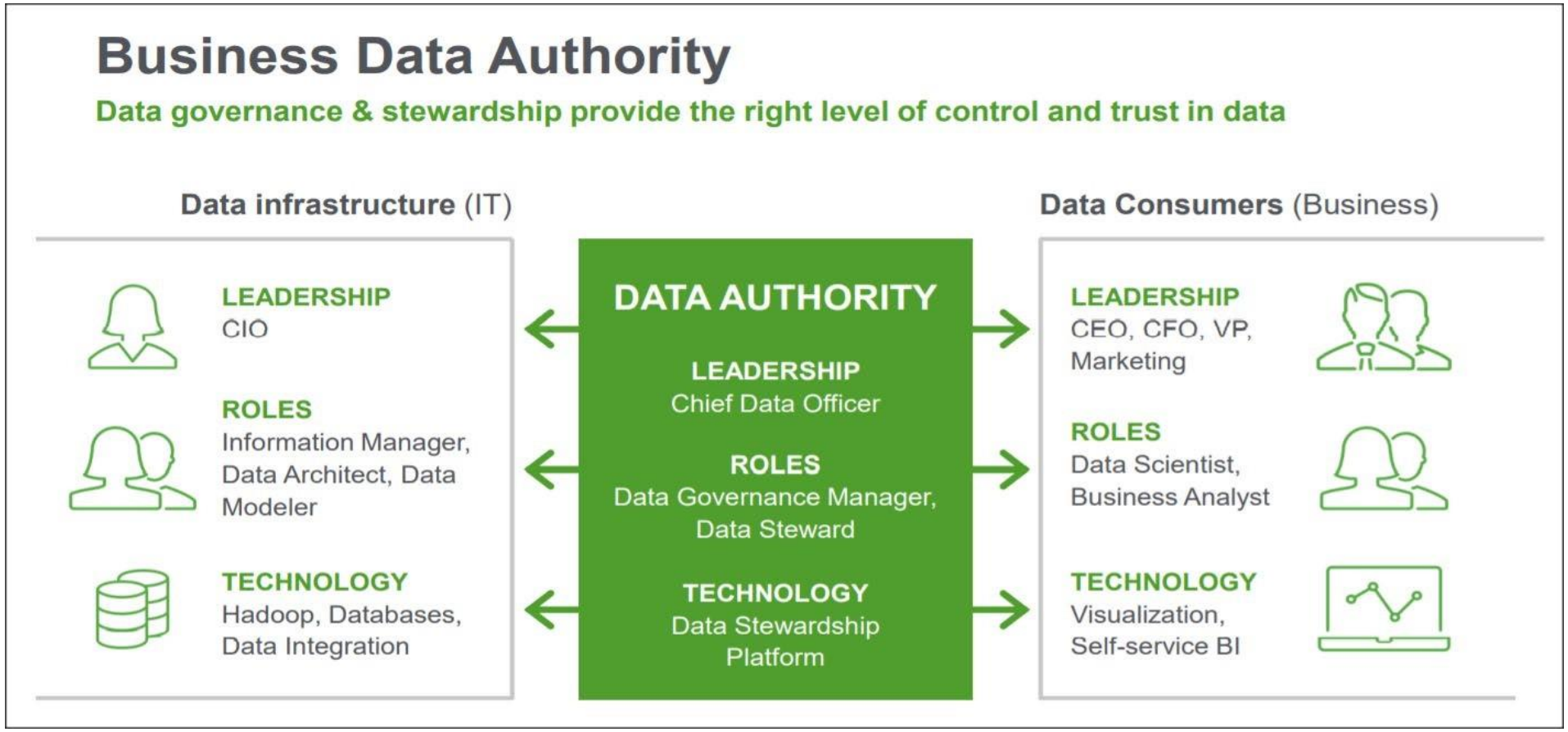
- Only few organizations have a CDO but industry analysis suggests increasing interest in the role especially in regulated sectors
- Research shows a range of functions for CDO
  - Establishing an organizational data strategy
  - Aligning data-centric requirements with available IT and business resources
  - Establishing data governance standards, policies and procedures
  - Providing advice (and perhaps services) to the business for data-dependent initiatives, such as business analytics, Big Data, data quality, and data technologies
  - Evangelizing the importance of good information management principles to internal and external business stakeholders
  - Oversight of data usage in analytics and Business Intelligence



- CDO (and their division) provides an interlocutor role between the data infrastructure functions (typically IT) and data consumers (typically business users)

Source: DATAVERSITY Launches Research Paper, "Status of the Chief Data Officer: An Update on the CDO Role in Organizations Today" - DATAVERSITY

CHIEF DATA OFFICER



Source: Collibra (2017) OVERVIEW OF DMBOK V2

DATA STEWARDSHIP (1/3)

Every company has data stewards, they may not be formalized

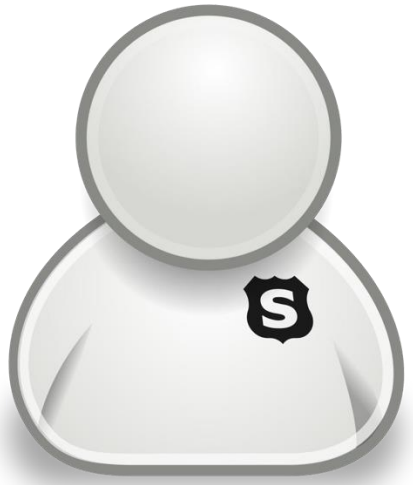
Data Stewardship: The process of having data stewards work with the data and metadata of an org to ensure its quality, accuracy, formats, domain values, and that is properly defined and understood across the enterprise.

Data Stewardship's role is to ensure organizational data and metadata meet quality, accuracy, format and value criteria; ensuring that data is properly defined and understood (standarized) across the enterprise

Source: J. Ladley

DATA STEWARDSHIP (2/3)

- *A data steward is a role, not a job.*
- Data Stewards represent the interests of all stakeholders to ensure enterprise data is of high quality and can be used effectively.
- Effective Data Stewards are accountable and responsible for data governance activities and have a portion of their time dedicated to these activities.
- Data steward roles activities will change depending on organisational circumstances but may include:
  - **Creating and managing core metadata** such as business terminology, data dictionary.
  - **Documenting rules and standards** in order to have organisational consensus and consistent use.
  - **Managing data quality issues** such as their identification, analysis and resolution (e.g., data integration)
  - **Executing operational data governance activities** such as participating in data management projects, data governance monitoring
  - **Ensuring compliance** with data standards
  - **Facilitates communication** and coordination **between business and IT**

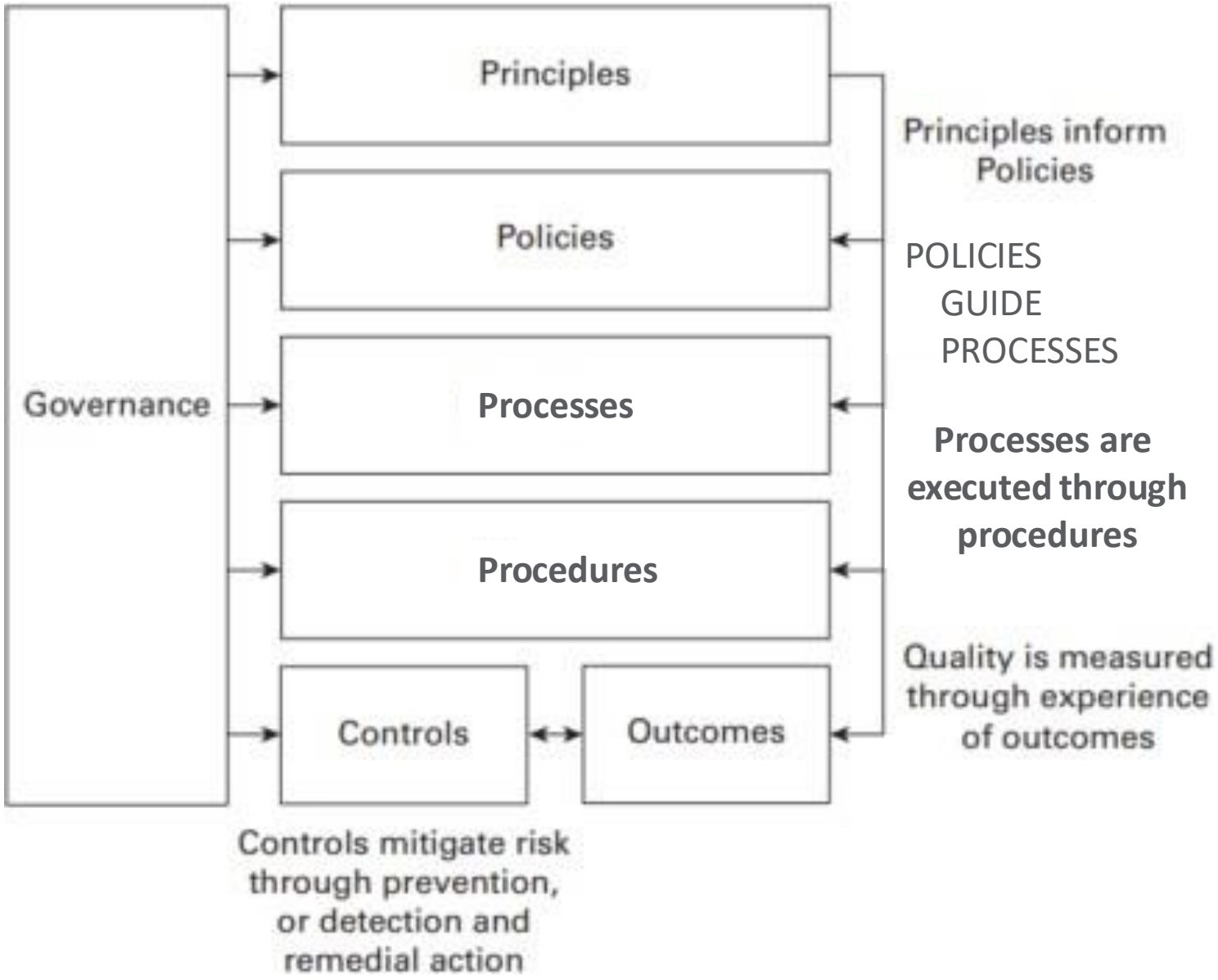


DATA STEWARDSHIP (3/3)

Executive Data Stewards	Senior managers who serve on a Data Governance Council.
Enterprise Data Stewards	Have oversight of a data domain across business functions.
Business Data Stewards	Business professionals - subject matter experts - accountable for a subset of data. They work with stakeholders to define and control data.
Data Owner	A business Data Steward, who has approval authority for decisions about data within their domain.
Technical Data Stewards	IT professionals operating within one of the Knowledge Areas, such as Data Integration Specialists, Database Administrators, Business Intelligence Specialists, Data Quality Analysts or Metadata Administrators.
Coordinating Data Stewards	Lead and represent teams of business and technical Data Stewards in discussions across teams and with executive Data Stewards. Coordinating Data Stewards are particularly important in large organizations.


Management of
principles,
policies,
processes, and
procedures

The hierarchy of principles to outcomes



Book: O'Keefe, K. and Brien, D.O., 2018. Ethical data and information management: concepts, tools and methods. Kogan Page Publishers.

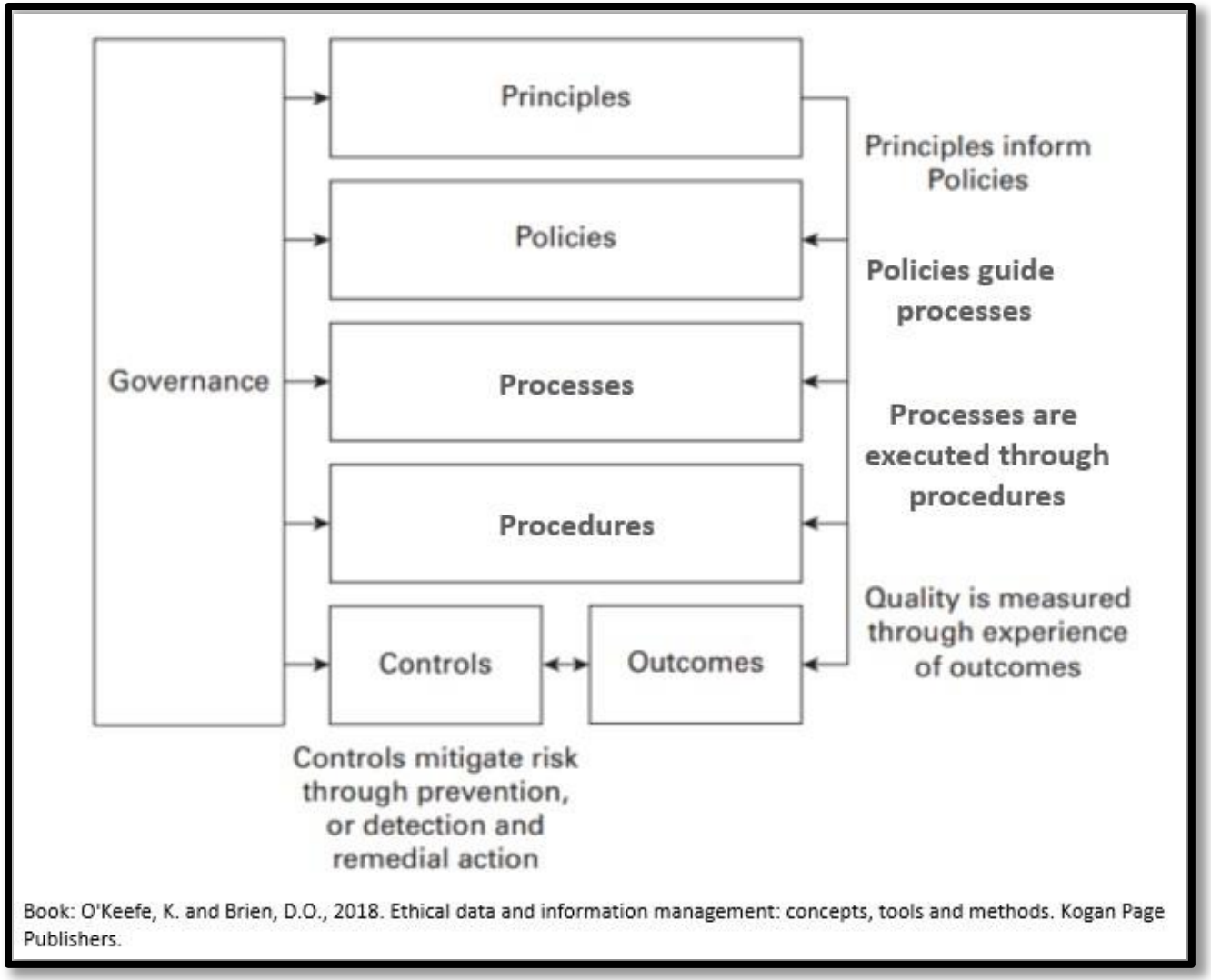
DATA POLICIES, PROCESSES, AND PROCEDURES

Definitions

- **Principles:** "**statements of philosophy** (or **core beliefs**) around policies and behaviours around data governance to be used as guidance for procedures and decision-making efforts" (Ladley)
- **Policy:** "a course or **principle of action** adopted or proposed by an organisation or individual."
- **Process:** "**a series of actions or steps** taken in order to achieve a particular end."
- **Procedure:** "an established or official **way of doing something**."

More intuitively:

- **Principles:** "a **fundamental law**, doctrine, premise, or assumption, a **code of conduct**"
- **Policy:** "is a **rule, regulation**, or **set of guidelines**"
- **Process:** "is a **high-level set of things** that must happen outlining what must happen in order to ensure compliance with a policy."
- **Procedure:** "is a specific, **detailed series of actions** that staff members must take in order to implement a process and comply with a policy"



Book: O'Keefe, K. and Brien, D.O., 2018. Ethical data and information management: concepts, tools and methods. Kogan Page Publishers.

POLICIES, PROCESSES, AND PROCEDURES

Example:

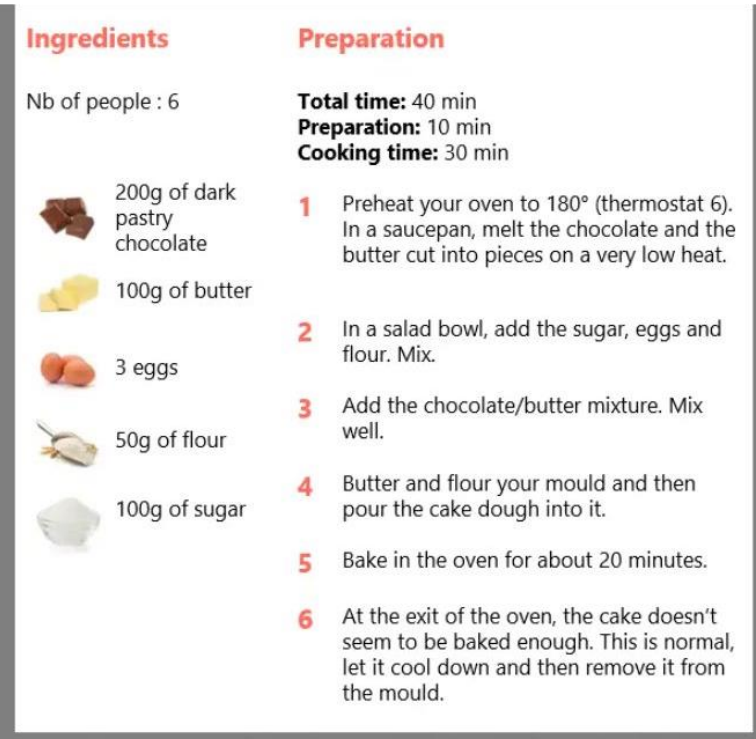
•A **policy** establishes the rules of being a bakery, such as proper use of utensils or appliances for safety purposes.

•A **process** explains what the bakery makes and sells (i.e., process for making a chocolate cake) and who is involved at each level (e.g., hiring a new chef).

•A **procedure** is the recipe for a cake, giving you the exact ingredients and steps to make the cake; the exact procedure to recruiting a new chef



"Making a Chocolate Cake" Process



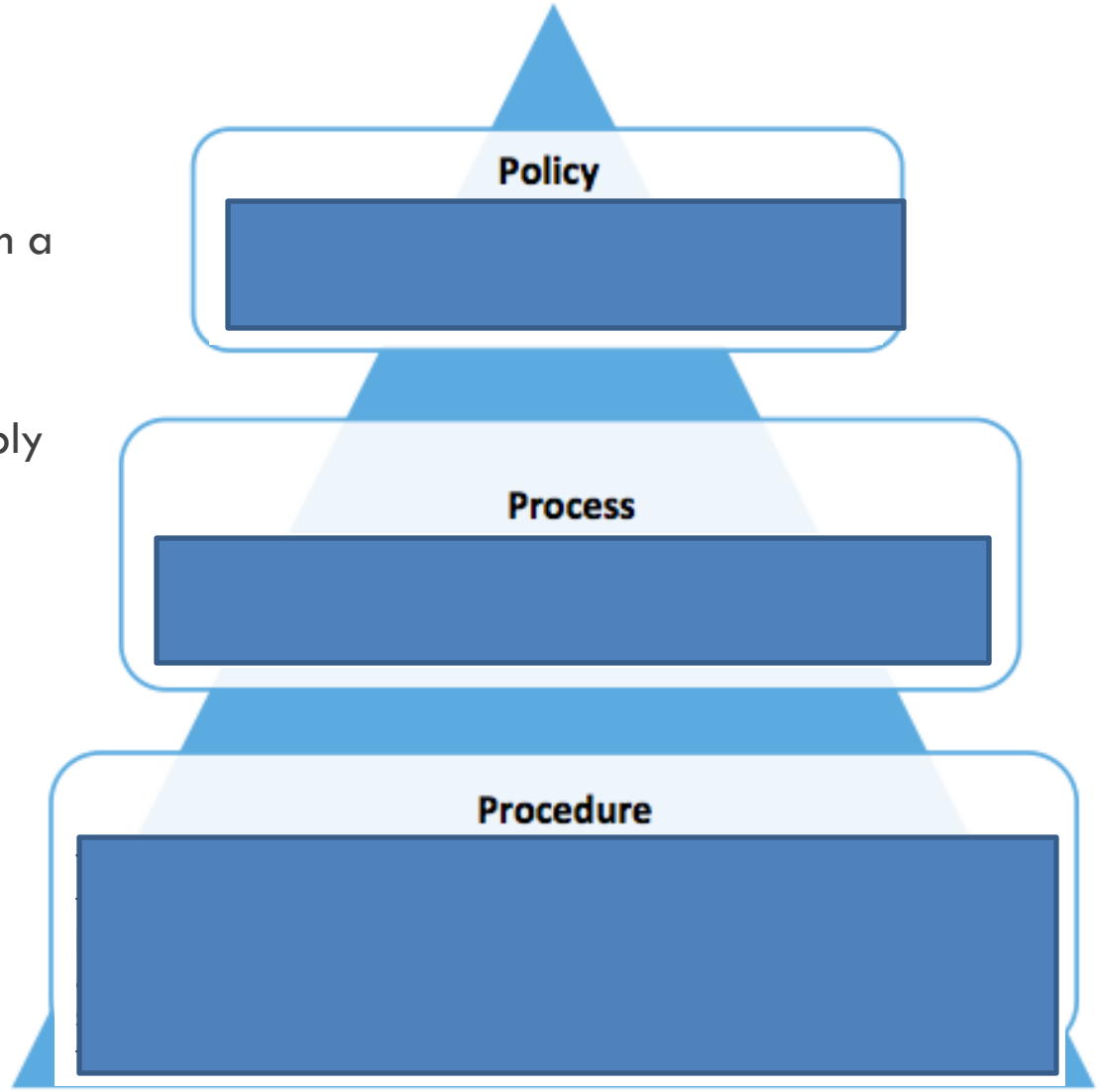
Paper procedure "Making a Chocolate Cake".



DATA POLICIES, PROCESSES, AND PROCEDURES

More intuitively:

- **Policy**: "is a rule, regulation, or **set of guidelines**"
- **Process**: "is a **high-level set of things** that must happen outlining what must happen in order to ensure compliance with a policy."
- **Procedure**: "is a specific, **detailed series of actions** that staff members must take in order to implement a process and comply with a policy"



Example #1 (Online Shop): A company promises that a product can be delivered to a customer within 24 hours of when the customer places an order.

Example #2 (School Fire Drill): Fire departments require schools to hold fire drills every month.

DATA GOVERNANCE POLICY

A data governance policy is a documented **set of guidelines** for ensuring that an organization's data and information assets are managed consistently and used properly.

A data policy describe the "what" of data governance (what to do, what not to do); while **procedures describe "how"** to do data governance.

Data policies can vary widely across organizations

Examples:

- ✓Sensitive data access and handling policies
- ✓Data Usage policies
- ✓Data Classification policies
- ✓Organizational ownership policies



- Handling requirements for data daily use
- Approvals required for sharing PII data with vendor or 3rd parties
- Data masking procedures
- etc.

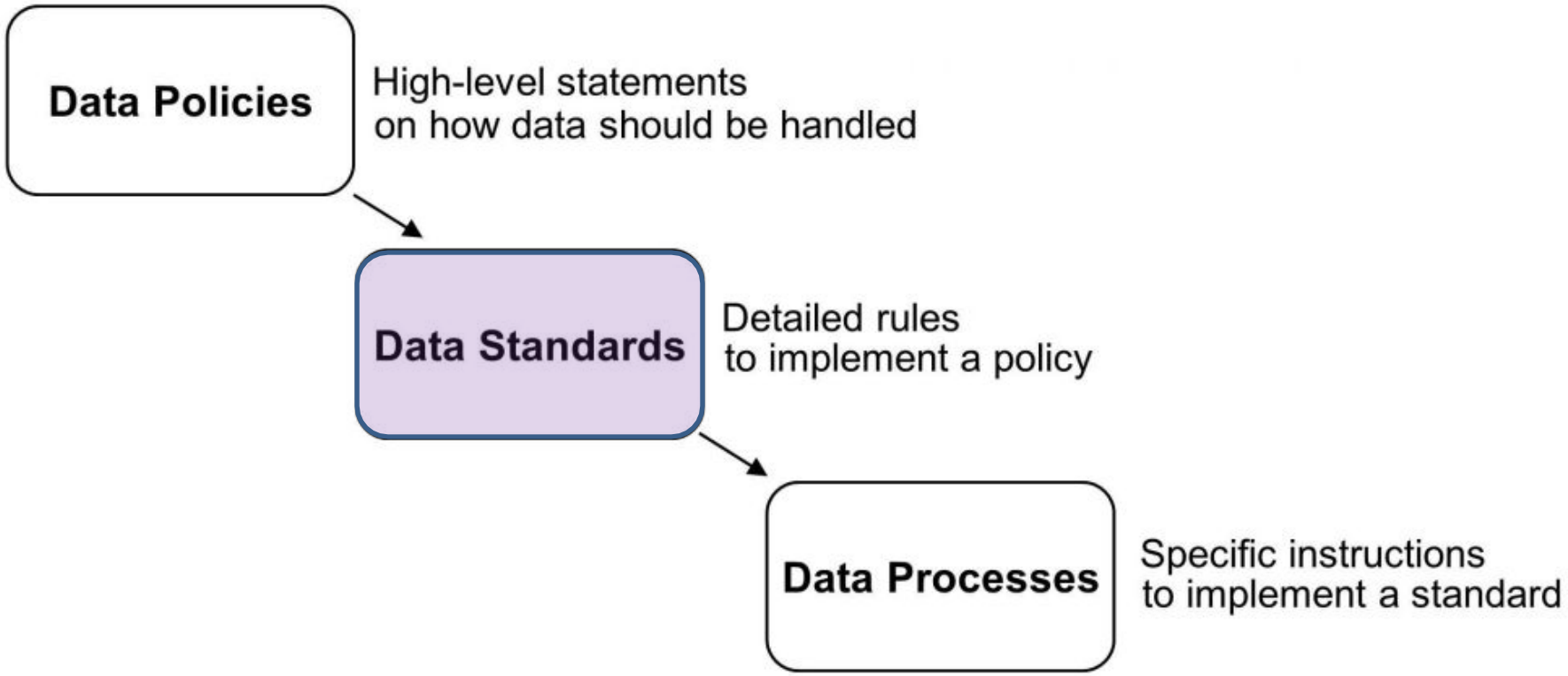
DATA GOVERNANCE POLICY STRUCTURE



A data governance policy typically include the following:

1) the **inventory of the data sources** within the organization;
2) the **goals** of the organization's data governance program **and metrics for** determining **success;**
3) the **positions** within the organization that will oversee elements of the governance program;
4) expectations around **data quality and data lifecycle management** as well as expectations around **data integrity and data integration;**
5) information about **data access** (parameters to establish which roles can access which data elements);
6) details around acceptable **data usage;**
7) different **categories of data** based on whether it's sensitive, confidential or publicly available, along with the levels of security and protection required at the different levels;
8) the **laws and regulations** that must be followed and what compliance means for the organization's data program.

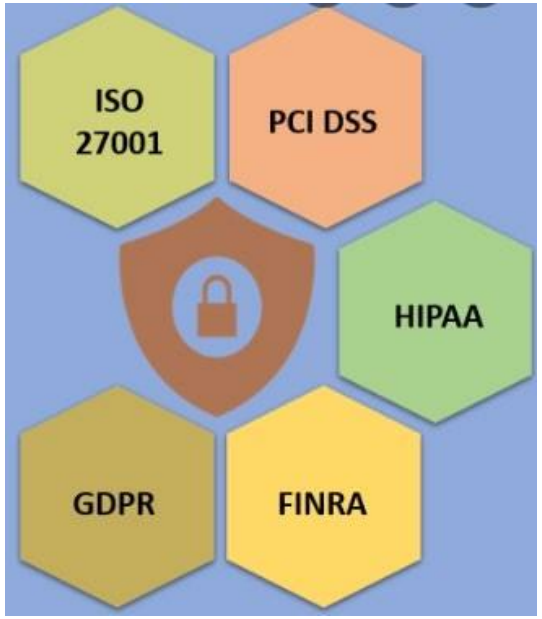
DATA STANDARDS



STANDARDS

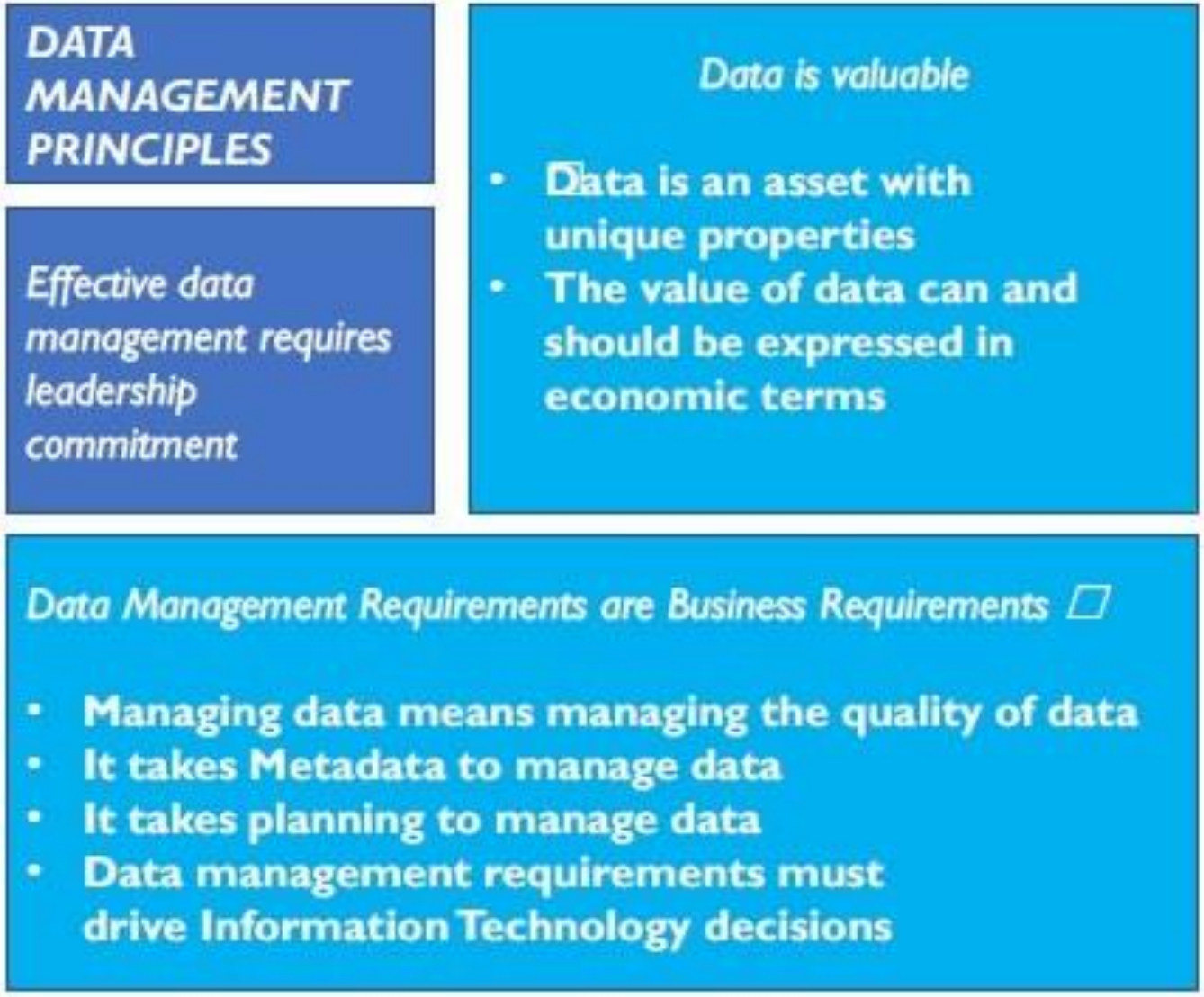
- A standard is defined as “something set up and established by authority as a rule for the measure of quantity, weight, extent, value, or quality.”
- By adopting a standard, an organization makes a decision once and codifies it in a set of assertions (the standard).
- DG standards should be mandatory.
- Drafted by data management professionals but reviewed, approved and adopted by the Data Governance Council.
- Data standards must be effectively communicated, monitored, and periodically reviewed and updated.

- Examples of data standards might include:
  - Data Architecture: Enterprise data models, tool standards, and system naming conventions
  - Acceptable use policies
  - Data Quality standards (timeliness, accuracy)
  - Data security standards (PCI DSS, ISO 27001)
  - Use of data for data analytics and presentation purposes



DATA MANAGEMENT PRINCIPLES

- DG shares characteristics with other forms of asset management
- Strategic and operational needs must be balanced.
- Striking a balance requires to follow Data Management principles, which may include:



DMBOK Figure 1 Data Management Principles

DATA STRATEGY

A strategy is "a set of choices and decisions that together chart a high-level course of action to achieve high-level goals. …
A strategic plan is a high-level course of action to achieve high-level goals." (DMBOK)

"Data strategy must come from an understanding of the data needs inherent in the business strategy:

- ✓ what data the organization needs
- ✓ how it will get the data
- ✓ how it will manage it and ensure its reliability over time, and
- ✓ how it will utilize it."

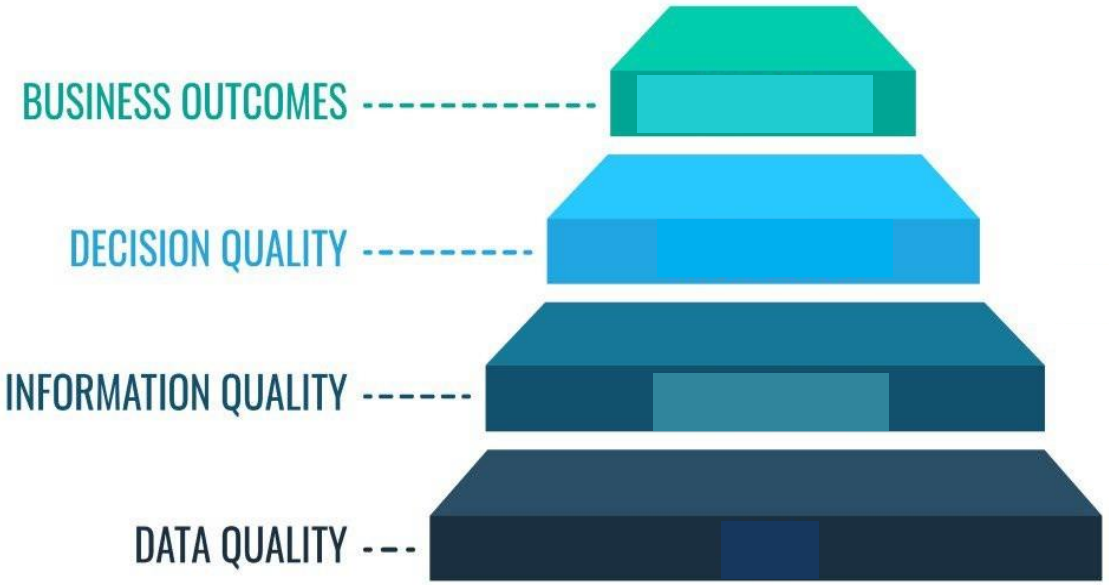




Data Quality

DATA QUALITY

- Data quality – refers to ensuring data is correct, consistent, and free of "noise" that might impeded usage and analysis.



The DIKW pyramid

Example

Applied: I'd better stop the car

Context: The traffic light I am driving towards has turned red

Meaning: South facing traffic light on corner of O'Connell St and Abbey St has turned red

Raw: Red color

DATA QUALITY VS. DATA GOVERNANCE



is Data Governance the same as Data Quality?

- **Data Quality** – The degree to which data is accurate, complete, timely, and consistent with all requirements and business rules.

- **Data Governance** – The exercise of authority, control, and shared decision making (e.g. planning, monitoring, and enforcement) over the management of data assets.

MEASURING DATA QUALITY (1/4)

- A quantitative measure are the data quality Key Performance Indicators (Data quality KPIs).
- Metrics help to differentiate between high-quality data and low-quality data.
- Data quality is generally measured against six 'data quality dimensions'



- ✓ Completeness
- ✓ Uniqueness
- ✓ Timeliness
- ✓ Validity
- ✓ Accuracy
- ✓ Consistency

MEASURING DATA QUALITY (2/4)

1. **Completeness**

Measures whether all the necessary data is present in a specific dataset.

Ex: survey



2. **Uniqueness**

Assesses how unique a data entry is, and whether it is duplicated anywhere else within your database. Ex: DB redundancy



MEASURING DATA QUALITY

3. **Timeliness**

Assesses how useful or relevant your data may be based on its age. Ex: car mileage, Netflix recommender system



4. **Validity**

Assesses how well data conforms to required value attributes.
Ex: format validation



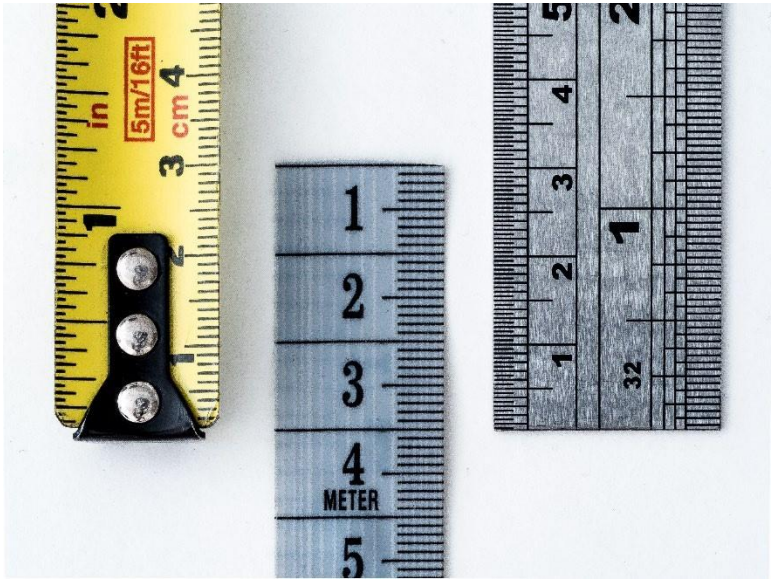
MEASURING DATA QUALITY

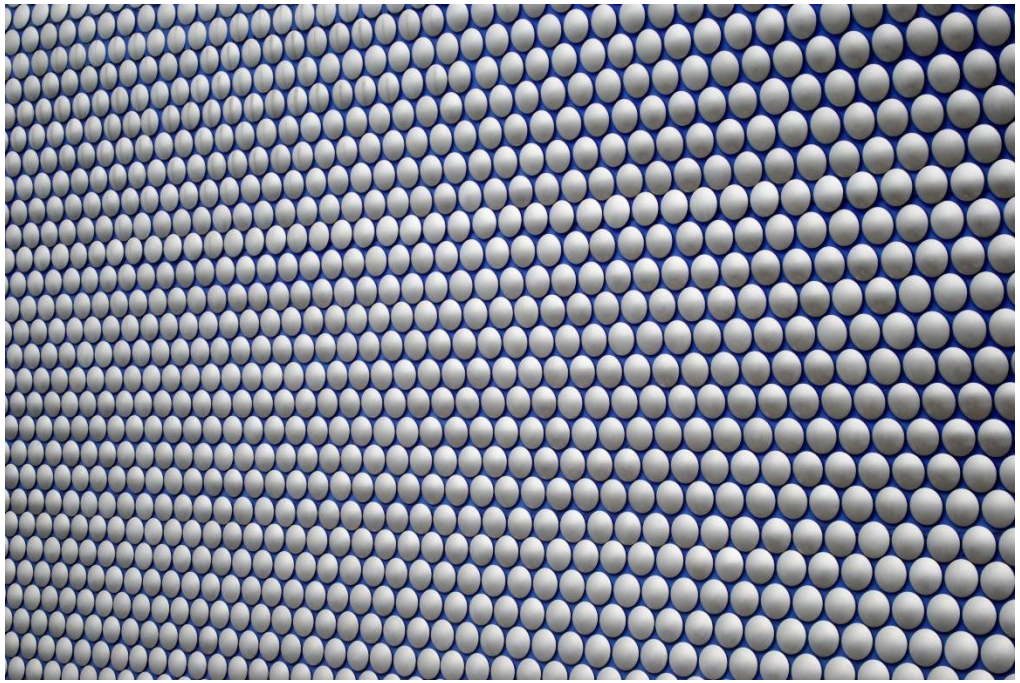
5. **Accuracy**

Determines whether the information you hold is correct or not (data correctly describes the real world objects being described). Ex: Bank account balance

6. **Consistency**

Refers to maintaining synchronicity between different databases.
Ex: info in a department store



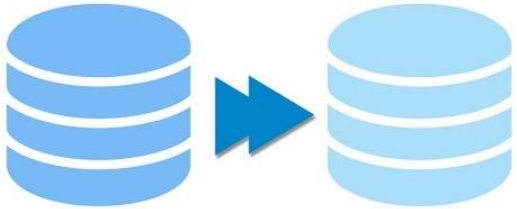


REASONS TO CARE ABOUT DATA PROVENANCE

1. To improve/track/ensure data quality
2. To provide an audit trail
3. To aid replication
4. To support attribution
5. Informational







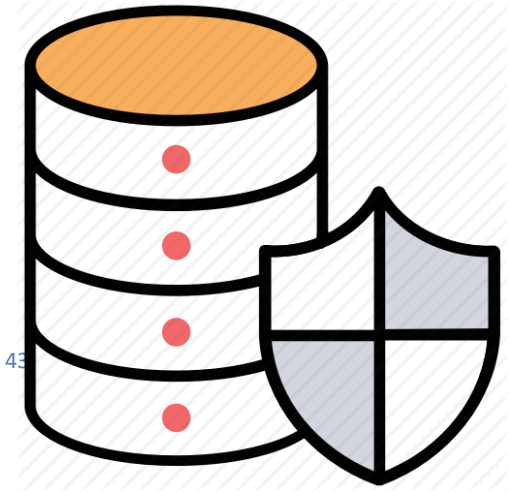





Data Security
and
Integrity

DATA INTEGRITY

- **Data integrity** is the maintenance of, and the **assurance of the accuracy, completeness, and consistency of data** over its entire life-cycle
- It is a critical aspect to the design, implementation, and usage of any system which stores, processes, or retrieves data
- It is maintained by a collection of processes, rules, and standards implemented during the design phase.
- Data integrity ensures that:
  - ✓ The information stored in a database remains complete, accurate, and reliable
  - ✓ Your data is safe from any outside forces



TYPES OF DATA INTEGRITY

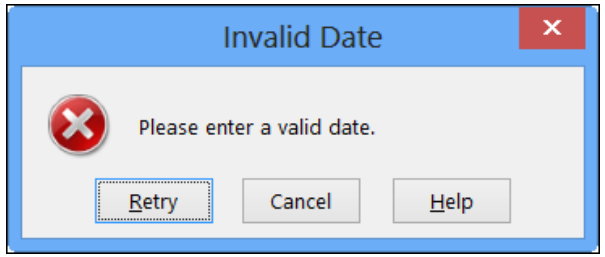
1. **Physical integrity** is the protection of data's wholeness and accuracy as it's stored and retrieved. Compromised when natural disasters strike, power goes out, or hackers disrupt database functions.

2. **Logical integrity** keeps data unchanged as it's used in different ways in a relational database.
   - **Entity integrity** ensures that **data is not listed more than once** and that no field in a table is null.
   - **Referential integrity** makes sure data is stored and used uniformly. Ensures that **only appropriate changes, additions, or deletions of data occur**.
   - **Domain integrity** ensures the accuracy of each piece of data in a domain. It can include constraints and other measures **that limit the format, type, and amount of data entered**.
   - **User-defined integrity** involves the rules and constraints created by the user to fit their **particular needs**.

RISK TO DATA INTEGRITY

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Domain integrity



1. Date Outside An Acceptable Range

Domain integrity

2. Phone Number In The Wrong Format

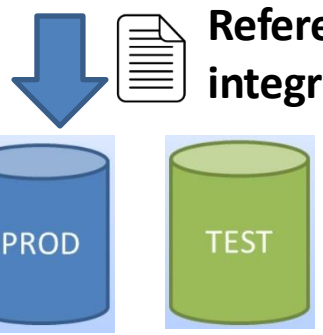


Referential integrity



3. Delete the wrong record or insert data into the wrong table

4.



5. Enter test data directly into the production database

Entity integrity

contact_id	first_name	last_name	email
1	Syed	Abbas	syed.abbas@example.com
2	Catherine	Abel	catherine.abel@example.com
3	Kim	Abercrombie	kim.abercrombie@example.com
4	Kim	Abercrombie	kim.abercrombie@example.com
5	Kim	Abercrombie	kim.abercrombie@example.com
6	Hazem	Abolrous	hazem.abolrous@example.com
7	Hazem	Abolrous	hazem.abolrous@example.com
8	Humberto	Acevedo	humberto.acevedo@example.com
9	Humberto	Acevedo	humberto.acevedo@example.com
10	Pilar	Ackerman	pilar.ackerman@example.com

8.Allow to insert duplicate records

Physical integrity



7. Fire sweeps and burns the database computer

Physical integrity



6. Hacker deletes all the database

DATA INTEGRITY CHALLENGES

1. Data Comes From Various Sources



2. Lack Of Data Integrity Policy



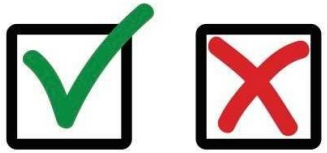
3. Lack Of Quality Assurance (QA) And Quality Control (QC) Review



4. Lack Of Data Governance System



5. Lack Of Validation Process



DATA SECURITY

Difference Between Data Security And Data Integrity

- Data security is the protection of data against unauthorized access or corruption and is necessary to ensure data integrity.
- Data integrity is a desired result of data security, but the term data integrity refers only to the validity and accuracy of data rather than the act of protecting data.



End of Week 2

QUESTIONS?